

Assume participation decreases with age but mean and std stay the same

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Set random seed for reproducibility
np.random.seed(42)

# Generate data for running events
ages = [5*x for x in range(2,20)]

participation = [10000/(x*np.sqrt(x)) for x in ages]

# performances = [np.max(np.random.normal(50,20,p)) for p in participation]

performances = []
for i,p in enumerate(participation):
    performances.append(np.max(np.random.normal(50,10,[int(p)])))

plt.plot(participation, performances)
plt.gca().invert_xaxis()
plt.xlabel("Participation")          # Label for the X-axis
plt.ylabel("Performance")          # Label for the Y-axis
plt.title("Performance decreasing with participation (participation decreases with age)") #
plt.show()
```

Performance decreasing with participation (participation decreases with age)

